Customer Segment & Chips Purchasing Behaviour Analysis

Strategic Recommendation Report for Julia, Category Manager

1. Executive Summary

This report analyses customer purchasing behaviour within the chips category using transaction and customer segment data. The objective is to identify which segments drive sales, which brands perform strongest, what influences spending behaviour, and where commercial growth opportunities exist.

Key Outcomes
Kettle is the highest-performing brand by revenue (390K+), with Mainstream Young Singles/Couples showing a brand affinity score of 1.20 means 20% of these segment shows interest towards kettle.
Older Singles/Couples generate the highest overall category revenue (0.4M).
Older Families deliver the highest average spend per customer (36) driven by high purchase volume, not price.
Young Singles/Couples (Mainstream) represent the largest customer base (8,088 customers) and a high-growth revenue opportunity.
175g packs are the strongest-selling pack size across all segments.

2. Business Objective

Provide Julia with data-backed recommendations for the upcoming category review by understanding current customer behaviour and revenue drivers.

Focus Area	Description
Revenue by Segment	Identify which lifestage and premium segments drive total sales
Brand Preference	Determine brand affinity scores by segment to guide shelf strategy
Pack Size	Find preferred pack sizes per segment for inventory prioritisation
Spend per Customer	Understand avg spend to identify high-value vs high-volume segments
Growth Opportunities	Recommend specific commercial actions backed by quantified revenue impact

3. Datasets Used

Dataset	Key Columns
Transaction Data	Date, Store Number, Customer ID, Product Name, Quantity, Total Sales
Purchase Behaviour	Customer ID, LIFESTAGE, PREMIUM_CUSTOMER

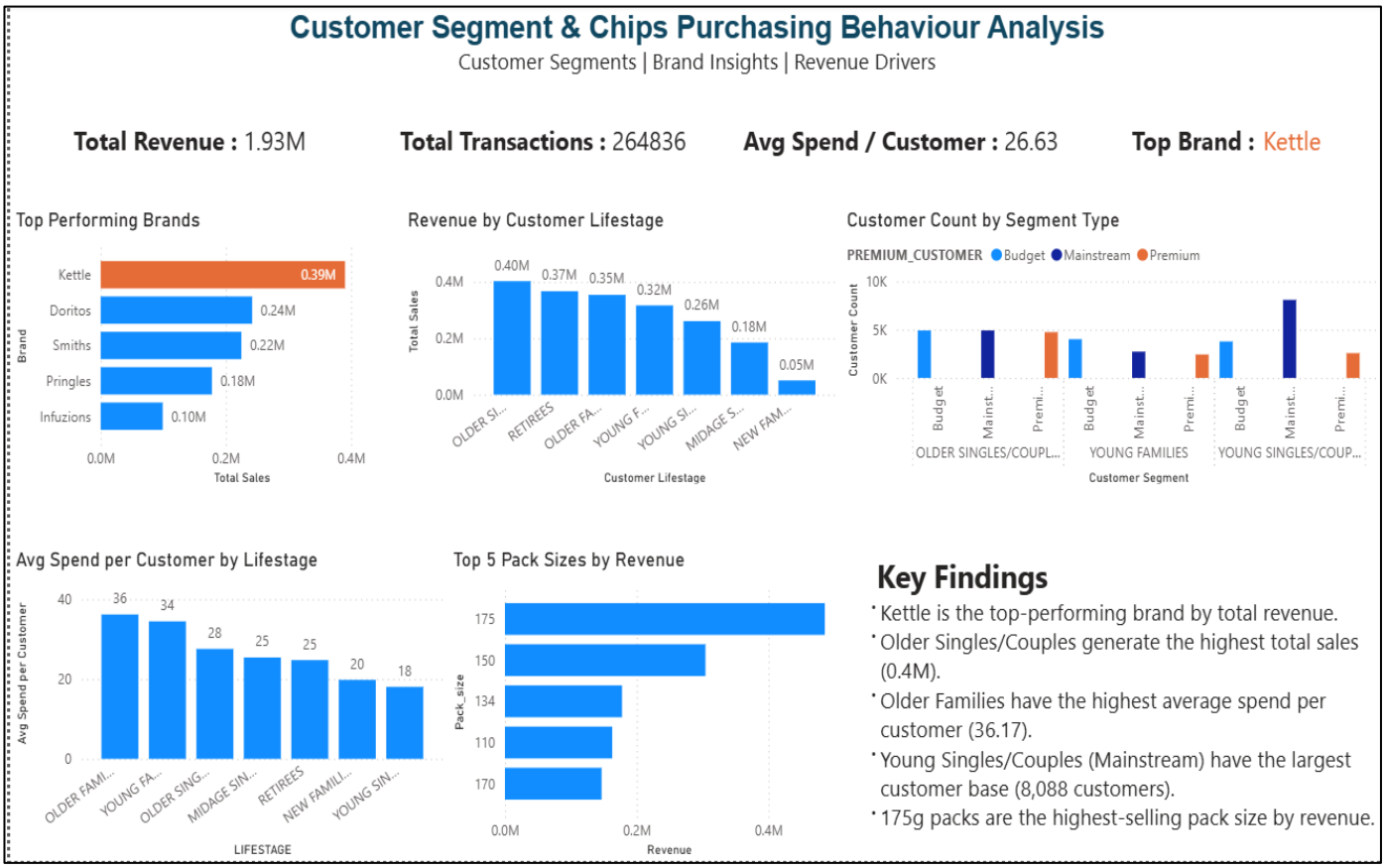
4. Data Preparation & Cleaning

1. Imported raw CSV files into MySQL
2. Converted Excel serial dates into standard SQL date format
3. Cleaned hidden characters and inconsistent text values
4. Extracted PACK_SIZE and BRAND from PROD_NAME using REGEXP
5. Joined transaction and purchase behaviour tables on LYLTY_CARD_NBR

5. Analytical Methods Used

SQL Techniques	Business Analysis Methods
INNER JOIN across tables	Segment revenue comparison
GROUP BY with aggregations (SUM, AVG, COUNT DISTINCT)	Brand affinity scoring
Subqueries for avg spend per customer	Pack size preference analysis
CASE WHEN for conditional aggregation	Customer value vs volume distinction
REGEXP for feature extraction from text	Impulse vs planned purchase identification

6. Dashboard



7. Key Findings

Finding 1: Kettle is the Leading Brand — Supported by Brand Affinity Analysis

Metric	Value
Kettle Total Revenue	390K+
Doritos Total Revenue	241K+
Smiths Total Revenue	224K+
Kettle Affinity — Mainstream Young Singles	1.20 (20% more likely to buy Kettle)
Doritos Affinity — Mainstream Young Singles	1.15 (15% more likely to buy Doritos)
Smiths Affinity — Mainstream Young Singles	0.77 (23% LESS likely to buy Smiths)

Business Meaning: Kettle dominates both total revenue AND segment-level preference. Smiths affinity below 1.0 for this segment indicates shelf space for Smiths targeting young singles may be inefficient. Doritos is a strong secondary brand for this segment.

Finding 2: Older Singles/Couples Drive Highest Total Revenue

Metric	Value
Total Revenue	0.40M (highest of all segments)
Driver	High transaction frequency, not just pack size

Business Meaning: This is the most commercially valuable segment by total contribution. Retention strategy is critical — losing these customers has the highest revenue impact.

Finding 3: Older Families Have Highest Spend Per Customer

Metric	Value
Avg Spend / Customer	36 (highest of all segments)
Avg Quantity / Customer	~2.89 units
Avg Price / Unit	7.09
Driver	High volume purchases, not higher price per unit

Business Meaning: Older families buy more units per visit. Multi-pack and family-size promotions are highly relevant for this segment.

Finding 4: Young Singles/Couples (Mainstream) — High Growth Segment

Metric	Value
Customer Count	8,088 (largest customer base)
Avg Spend / Customer	18
Current Total Revenue	1,45,584
Kettle Brand Affinity	1.20 — 20% more likely to buy Kettle than population avg
Doritos Brand Affinity	1.15 — 15% more likely to buy Doritos
Smiths Brand Affinity	0.77 — actively avoids Smiths

Kettle 175g Sales by this segment	64,121.5 (8,953 packets)
Revenue if avg spend rises to 22	1,77,936 (+32,352 uplift)

Business Meaning: Large customer base + clear brand preference = scalable, targeted growth opportunity. Every Rs.4 increase in avg spend = Rs.32,352 additional revenue.

Finding 5: 175g Packs Lead Category Revenue

Pack Size	Revenue
175g	Highest
150g	Second
134g	Third

Business Meaning: 175g should receive strongest shelf visibility and inventory priority. This is the pack size Kettle should be stocked in most heavily.

8. Strategic Recommendations

Recommendation 1: Target Mainstream Young Singles with Kettle 175g	
Finding	Mainstream Young Singles have a Kettle affinity score of 1.20, meaning they are 20% more likely to buy Kettle than the rest of the population. Kettle 175g generates 64,121 in sales from this segment with 8,953 packets sold.
So What	8,088 customers currently spend an average of 18. Increasing average spend to 22 generates an additional 32,352 in revenue from this segment alone — no new customer acquisition needed.
Action	1. Place Kettle 175g in high-visibility offer zones and near checkout for impulse purchase. 2. Run buy-2-get-10%-off promotion specifically on Kettle 175g. 3. Avoid promoting Smiths to this segment (affinity 0.77 — they actively avoid it).
Expected Outcome	Increased purchase frequency and basket size for the largest customer base. Rs.4 avg spend increase = Rs.32,352 additional revenue. Scalable across stores.

Recommendation 2: Retain and Grow Older Singles/Couples	
Finding	Older Singles/Couples generate the highest total revenue of 0.40M driven by high transaction frequency. They are the most commercially critical segment to retain.
So What	Losing even 10% of this segment removes 40,000 in annual category revenue. No other segment can compensate at current spend rates.
Action	1. Ensure consistent Kettle stock availability — stockouts directly hurt this segment. 2. Offer loyalty-based multi-buy deals to reward purchase frequency. 3. Position Kettle at eye level in aisles frequented by this demographic.
Expected Outcome	Retained revenue base of 0.40M. Frequency maintenance = category stability.

Recommendation 3: Grow Older Families Through Volume Promotions	
Finding	Older Families have the highest average spend per customer (36) driven by buying 2.89 units per visit at a price of 7.09 per unit.
So What	4,675 customers spending 36 avg = 1,68,300 total. A 10% customer base increase (467 new customers) adds 16,812 in revenue.
Action	1. Promote family-size and multi-pack bundles specifically to this segment. 2. Place larger pack sizes (300g, 380g) at family shopping aisle locations. 3. Use value messaging — these are budget customers who respond to volume deals.
Expected Outcome	Customer base growth of 10% = 16,812 additional revenue from a single segment.

Recommendation 4: Optimise Pack Size Mix — Lead with 175g	
Finding	175g is the top-selling pack size by revenue across all segments. Kettle 175g is the #1 SKU combination.
So What	Suboptimal shelf space allocation to low-performing sizes reduces category revenue per shelf metre.
Action	1. Allocate maximum shelf facings to 175g packs — especially Kettle 175g. 2. Review and reduce shelf space for sizes below 4,000 units (90g, 125g, 220g). 3. Ensure 175g is never out of stock on weekends — peak transaction period.
Expected Outcome	Improved category revenue per shelf metre. Reduced stockout risk on #1 SKU.

Recommendation 5: Use Secondary Brands Strategically by Segment	
Finding	Doritos affinity for Mainstream Young Singles = 1.15. Smiths affinity = 0.77. Brand preference is highly segment-specific.
So What	Treating all brands equally across segments wastes promotional budget. Doritos is a strong

	secondary brand for young segments. Smiths performs better in older segments.
Action	1. Co-promote Kettle + Doritos for young singles (both have affinity above 1.0). 2. Allocate Smiths promotions toward older family segments where affinity is higher. 3. Use affinity scores to guide all future promotional planning by store demographic.
Expected Outcome	Higher promotional ROI by matching brand investment to segment preference.

9. Brand Affinity Analysis — Mainstream Young Singles/Couples

Brand affinity measures how much more (or less) likely a specific segment is to purchase a brand compared to the rest of the population. Formula: (Brand % of segment purchases) / (Brand % of all other purchases). Score above 1.0 = segment over-indexes on this brand. Below 1.0 = under-indexes.

Brand	Segment Kettle %	Population %	Affinity Score	Interpretation
Kettle	18.56%	15.50%	**1.20**	20% MORE likely — prioritise
Doritos	12.28%	10.65%	**1.15**	15% MORE likely — strong secondary
Smiths	9.34%	12.11%	**0.77**	23% LESS likely — avoid promoting to this segment

10. SQL Queries Used

A. Data Preparation — Date Conversion

```
SET SQL_SAFE_UPDATES = 0;
ALTER TABLE transactions
ADD COLUMN txn_date DATE;
UPDATE transactions
SET txn_date = DATE_ADD('1899-12-30', INTERVAL EXCEL_DATES DAY);
SET SQL_SAFE_UPDATES = 1;
SELECT txn_date FROM transactions LIMIT 5;
```

B. Feature Engineering — Pack Size & Brand

```
ALTER TABLE transactions
ADD COLUMN pack_size_g INT;
UPDATE transactions
SET pack_size_g =
CAST(
REGEXP_SUBSTR(PROD_NAME, '[0-9]+(?=g)')
AS UNSIGNED
);
SELECT PROD_NAME, pack_size_g
FROM transactions
LIMIT 10;

ALTER TABLE transactions
ADD COLUMN brand VARCHAR(50);
UPDATE transactions
SET brand = TRIM(SUBSTRING_INDEX(PROD_NAME, ' ', 1));
SELECT PROD_NAME, brand
FROM transactions
LIMIT 10;
```

C. Customer Segment Revenue Analysis

```sql
SELECT p.LIFESTAGE,
       ROUND(SUM(t.TOT_SALES),2) AS total_sales
FROM purchase_behaviour p
JOIN transactions t
ON p.LYLTY_CARD_NBR = t.LYLTY_CARD_NBR
GROUP BY p.LIFESTAGE
ORDER BY total_sales DESC;
```

D. Average Customer Spend by Lifestage

```sql
SELECT LIFESTAGE,
       AVG(customer_total) AS avg_customer_spend
FROM (
    SELECT t.LYLTY_CARD_NBR,
           p.LIFESTAGE,
           SUM(t.TOT_SALES) AS customer_total
    FROM transactions t
    JOIN purchase_behaviour p
    ON t.LYLTY_CARD_NBR = p.LYLTY_CARD_NBR
    GROUP BY t.LYLTY_CARD_NBR, p.LIFESTAGE
) x
GROUP BY LIFESTAGE
ORDER BY avg_customer_spend desc;
```

E. Brand Affinity — Mainstream Young Singles

```sql
SELECT
  (SUM(CASE WHEN t.brand = "Doritos" THEN t.prod_qty ELSE 0 END)
  / SUM(t.prod_qty) )*100 AS kettle_pct
FROM transactions as t
join
purchase_behaviour as p
on t.LYLTY_CARD_NBR = p.LYLTY_CARD_NBR
where p.lifestage="YOUNG SINGLES/COUPLES" and
      p.premium_customer="Mainstream";
```

```sql
SELECT
  (SUM(CASE WHEN t.brand = "Doritos" THEN t.prod_qty ELSE 0 END)
  / SUM(t.prod_qty) )*100 AS other_kettle_pct
FROM transactions as t
join
purchase_behaviour as p
on t.LYLTY_CARD_NBR = p.LYLTY_CARD_NBR
where p.lifestage<>"YOUNG SINGLES/COUPLES" and
      p.premium_customer<>"Mainstream";
```

11. Final Recommendation to Julia

The chips category has three commercially distinct segments requiring different strategies. Brand affinity analysis provides Julia with precise, actionable direction rather than generic category advice.

Priority	Segment	Strategy	Revenue Opportunity
1	Mainstream Young Singles (8,088 customers)	Place Kettle 175g in offer zones + buy-2 promotion. Avoid Smiths promotions (affinity 0.77).	**+32,352 if avg spend rises from 18 to 22**
2	Older Singles/Couples (highest revenue segment)	Retention focus. Stock availability + loyalty promotions for Kettle.	**Protect 0.40M revenue base**
3	Older Families (highest avg spend/customer)	Family bundles and multi-pack promotions. Value messaging for budget buyers.	**+16,812 from 10% customer growth**
4	All Segments	Prioritise 175g shelf space. Kettle 175g is the #1 SKU — never out of stock.	**Category revenue per shelf metre improvement**

12. Conclusion

Customer segmentation strongly influences purchasing behaviour in the chips category. Kettle emerged as the dominant brand not only by total revenue but also by brand affinity analysis — Mainstream Young Singles are 20% more likely to buy Kettle than the average customer, making it the right brand to invest in for this high-volume segment.

Older Families and Older Singles/Couples represent the highest-value customers by spend per customer and total revenue respectively. Young Singles/Couples represent the highest-growth opportunity given their large customer base and clear, quantifiable revenue upside.

These findings directly support pricing, merchandising, promotional planning, and future category growth decisions. The brand affinity framework introduced in this analysis should be applied to all future segment-level recommendations.